MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 6 (Units 16, 17 & 18): Conic Sections, Reduction of 2nd-Degree Equations & System of Circles

Bismillahir Rahmanir Rahim · Md. Ataullah | Roll: 260503 | Reg: 1054600 | Dept. of EECE, PUST (18th Batch, 2025–26)

Module 6 Overview

Units Covered in Module 6:

- **Unit 16:** Identification of Conics and Reduction of General Second-Degree Equations to Standard Forms (Central Conics & Parabolas).
- **Unit 17:** Standard Equations and Properties of Parabola, Ellipse, and Hyperbola (Cartesian & Polar Forms).
- **Unit 18:** System of Circles (Orthogonality, Radical Axis, Radical Center, Coaxial System of Circles & Limiting Points).

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	Module 1
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	Module 2
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	Module 3
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	Module 4
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	Module 5
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	✓ **This file**
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	Module 7
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	Module 8

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1 Unit 16: Identification and Reduction of Second-Degree Equations

1.1 Classification of $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

With discriminant $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$:

- $ab - h^2 > 0 \implies$ **Ellipse** (Circle if $a = b$ and $h = 0$).
- $ab - h^2 < 0 \implies$ **Hyperbola** (Rectangular Hyperbola if $a + b = 0$).
- $ab - h^2 = 0 \implies$ **Parabola**.

1.2 Solved PUST Past Exam Problems on Reduction of Conics

PUST Past Exam Question 1 (Reduction of a Central Conic) [7 Marks]
Question: Reduce the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to its standard form. Identify the conic and find its center, lengths of axes, eccentricity, latus rectum, and equations of the axes.

Solution (Exam Style): Here, $a = 11,\ h = 12,\ b = 4,\ g = -10,\ f = -20,\ c = -5.$

$ab - h^2 = (11)(4) - 12^2 = 44 - 144 = -100 < 0$ [Hence the conic is a Hyperbola].

Step 1 (Center (α, β) and New Constant c'):

$$\frac{1}{2}\frac{\partial F}{\partial x} = 11\alpha + 12\beta - 10 = 0$$
$$\frac{1}{2}\frac{\partial F}{\partial y} = 12\alpha + 4\beta - 20 = 0 \implies 3\alpha + \beta = 5$$
$$\implies \alpha = 2, \qquad \beta = -1 \quad \text{[Center} = (2, -1)\text{]}$$
$$c' = g\alpha + f\beta + c = (-10)(2) + (-20)(-1) - 5 = -5.$$

Step 2 (Characteristic Roots λ_1, λ_2 & Standard Equation):

$$\lambda^2 - (a+b)\lambda + (ab - h^2) = 0$$
$$\implies \lambda^2 - 15\lambda - 100 = 0$$
$$\implies (\lambda - 20)(\lambda + 5) = 0 \implies \lambda_1 = 20, \quad \lambda_2 = -5$$
$$\implies 20X^2 - 5Y^2 - 5 = 0$$
$$\implies \frac{X^2}{1/4} - \frac{Y^2}{1} = 1 \quad \text{[Standard form with } A = \tfrac{1}{2},\ B = 1\text{]}.$$

Step 3 (Lengths of Axes, Eccentricity, Latus Rectum & Axis Equations):

Transverse axis length $= 2A = 1$, Conjugate axis length $= 2B = 2$,

Eccentricity, $e = \sqrt{1 + \dfrac{B^2}{A^2}} = \sqrt{1+4} = \sqrt{5}$, Latus rectum $= \dfrac{2B^2}{A} = 4$,

Transverse axis $(\lambda_1 = 20)$: $(11 - 20)(x - 2) + 12(y + 1) = 0 \implies 3x - 4y - 10 = 0$,

Conjugate axis $(\lambda_2 = -5)$: $(11 + 5)(x - 2) + 12(y + 1) = 0 \implies 4x + 3y - 5 = 0.$

PUST Past Exam Question 2 (Reduction of a Parabola when $ab - h^2 = 0$) [7 Marks]
Question: Reduce $9x^2 - 24xy + 16y^2 - 18x - 101y + 19 = 0$ to standard form, and find its vertex, axis, latus rectum, and directrix.

Solution (Exam Style): Since $9x^2 - 24xy + 16y^2 = (3x - 4y)^2$, $ab - h^2 = 0$ (**Parabola**).

$(3x - 4y + \lambda)^2 = (6\lambda + 18)x + (101 - 8\lambda)y + \lambda^2 - 19$

For perpendicular lines: $3(6\lambda + 18) - 4(101 - 8\lambda) = 0$

$$\implies 50\lambda = 350 \implies \lambda = 7$$
$$\implies (3x - 4y + 7)^2 = 15(4x + 3y + 2)$$
$$\implies \left(\frac{3x - 4y + 7}{5}\right)^2 = 3\left(\frac{4x + 3y + 2}{5}\right) \quad \text{[Dividing by } 5^2 = 25\text{]}$$
$$\implies Y^2 = 3X \quad \text{[Where } 4a = 3 \implies a = \frac{3}{4}\text{]}.$$

- **Latus Rectum:** $4a = 3$; **Axis ($Y = 0$):** $3x - 4y + 7 = 0$; **Tangent at Vertex ($X = 0$):** $4x + 3y + 2 = 0$.
- **Vertex ($X = 0, Y = 0$):** $\left(-\frac{29}{25}, \frac{22}{25}\right)$; **Directrix ($X = -3/4$):** $16x + 12y + 23 = 0$.

2 Unit 17: Standard Equations & Properties of Parabola, Ellipse & Hyperbola

Property	Parabola ($y^2 = 4ax$)	Ellipse ($\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$)	Hyperbola ($\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$)
Eccentricity e	$e = 1$	$e = \sqrt{1 - \frac{b^2}{a^2}} < 1$	$e = \sqrt{1 + \frac{b^2}{a^2}} > 1$
Vertices	$(0, 0)$	$(\pm a, 0)$	$(\pm a, 0)$
Foci	$(a, 0)$	$(\pm ae, 0)$	$(\pm ae, 0)$
Directrices	$x = -a$	$x = \pm\frac{a}{e}$	$x = \pm\frac{a}{e}$
Latus Rectum	$4a$	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Tangent at (x_1, y_1)	$yy_1 = 2a(x + x_1)$	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

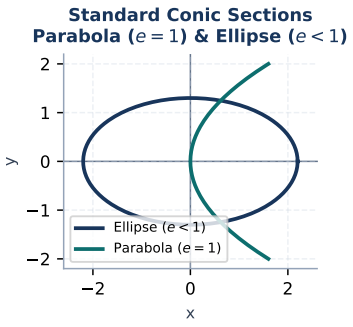
Polar Equation of a Conic (Focus at the Pole): $\frac{l}{r} = 1 + e\cos\theta$, where $l = ed$ is the semi-latus rectum.

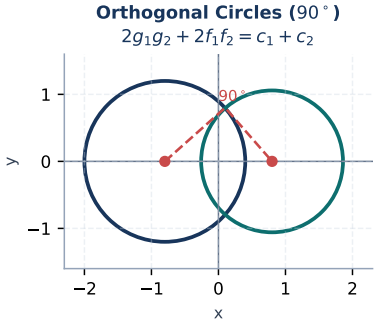
3 Unit 18: System of Circles

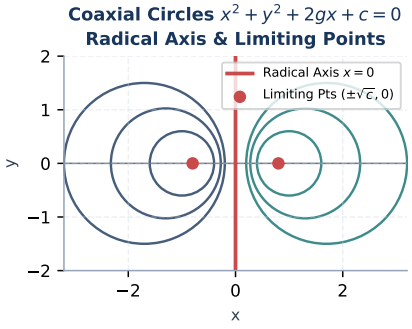
3.1 Orthogonality, Radical Axis, Coaxial System & Limiting Points

For two circles $S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$:

1. **Condition of Orthogonality:**
$$C_1C_2^2 = r_1^2 + r_2^2$$
$$\implies (-g_1 + g_2)^2 + (-f_1 + f_2)^2 = (g_1^2 + f_1^2 - c_1) + (g_2^2 + f_2^2 - c_2)$$
$$\implies 2g_1g_2 + 2f_1f_2 = c_1 + c_2.$$
2. **Radical Axis ($S_1 - S_2 = 0$):** $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$.
3. **Coaxial System of Circles:**
   - Canonical form: $x^2 + y^2 + 2gx + c = 0$ (g variable, c constant; radical axis $x = 0$; limiting points $(\pm\sqrt{c}, 0)$).
   - General family: $S_1 + \lambda(S_1 - S_2) = 0$ or $S_1 + \lambda S_2 = 0$ ($\lambda \neq -1$).







3.2 Solved PUST Past Exam Problems on System of Circles

PUST Past Exam Question 1 (Circle Orthogonal to Three Given Circles) [5 Marks]
Question: Find the equation of the circle that cuts orthogonally the three circles:

$$x^2+y^2+2x+17y+4=0, \qquad x^2+y^2+7x+6y+11=0, \qquad x^2+y^2-x+22y+3=0.$$

Solution (Exam Style): Let the required circle be $x^2+y^2+2gx+2fy+c=0$. Applying $2gg_i+2ff_i=c+c_i$ to each of the three given circles:

$$2g+17f-c=4 \tag{1}$$
$$7g+6f-c=11 \tag{2}$$
$$-g+22f-c=3 \tag{3}$$

Subtracting (1) from (2), and (3) from (1):

$$5g-11f=7$$
$$3g-5f=1$$
$$\implies g=-3, \qquad f=-2$$
$$\implies c=2(-3)+17(-2)-4=-44 \quad \text{[From equation (1)]}.$$

$\therefore$ The required orthogonal circle is $x^2+y^2-6x-4y-44=0$.

PUST Past Exam Question 2 (Limiting Points of a Coaxial System) [5 Marks]
Question: Find the radical axis and the limiting points of the coaxial system determined by $S_1\equiv x^2+y^2+2x+4y+7=0$ and $S_2\equiv x^2+y^2+4x+2y+5=0$.

Solution (Exam Style):

Radical axis $L\equiv S_1-S_2=0:\quad -2x+2y+2=0 \implies x-y-1=0$
Coaxial family $S_1+2\lambda L=0:\quad x^2+y^2+2(1+\lambda)x+2(2-\lambda)y+(7-2\lambda)=0$
Center $C=\big(-(1+\lambda),-(2-\lambda)\big)$
Radius $R^2=(1+\lambda)^2+(2-\lambda)^2-(7-2\lambda)=2\lambda^2-2.$

For limiting points, radius $R=0$:

$2\lambda^2-2=0 \implies \lambda=\pm1$
For $\lambda=+1:\quad C_1=(-2,-1)$
For $\lambda=-1:\quad C_2=(0,-3).$

$\therefore$ The limiting points are $(-2,-1)$ and $(0,-3)$.

Quick Revision — Module 6 (Units 16–18)
• **Conic Identification ($\Delta\neq0$):** $ab-h^2>0 \implies$ Ellipse; $ab-h^2<0 \implies$ Hyperbola; $ab-h^2=0 \implies$ Parabola.
• **Axis Equations of Central Conic:** $(a-\lambda_i)(x-\alpha)+h(y-\beta)=0$.
• **System of Circles:** Orthogonal condition: $2g_1g_2+2f_1f_2=c_1+c_2$; Radical axis: $S_1-S_2=0$; Limiting points: set radius $R=0$ in $S_1+\lambda L=0$.

Common Mistakes to Avoid in the Exam
• Trying to find the center (α,β) via $\frac{\partial F}{\partial x}=0, \frac{\partial F}{\partial y}=0$ when $ab-h^2=0$: a parabola has no center! Complete the square $(lx+my+\lambda)^2$ instead.